# Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Generative AI & Agentic Systems · Enterprise AI Deployment · RAG Pipelines · Responsible AI & Guardrails · Python · AWS · Azure · 9yr Production

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## PROFESSIONAL SUMMARY

AI Engineer with 9+ years of production Python and enterprise systems experience, now building and deploying generative AI and agentic solutions. Built LeaseI -- an 8-agent LangGraph autonomous pipeline with RAG, HITL escalation, and LLMOps evaluation harness (93.6% precision / 87.9% F1 across 1,240 compliance documents) -- and AgentProbe, an open-source AI evaluation toolkit with a CI-integrated 5-metric LLM-as-judge engine and multi-provider LLM routing. Enterprise deployment background: AWS (S3, EC2, RDS, Glue), Azure cloud services, 1M+ records at production scale, REST API integrations across enterprise systems. Responsible AI practitioner: code-enforced guardrails, HITL governance, and audit-trail design -- every compliance determination cited to source or explicitly flagged as unsupported. Work closely with technical and business stakeholders to translate complex problems into production-ready AI capabilities. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

## TECHNICAL SKILLS

**Generative AI & Agentic Systems:** LangChain, LangGraph, Multi-Agent Orchestration, Autonomous Agents, Workflow Automation, Tool Calling / Function Calling, Generative AI, Large Language Models, Prompt Engineering, Context Engineering, Few-Shot Prompting, Chain-of-Thought Prompting, Output Validation, HITL / Human-in-the-Loop, AI Guardrails, Responsible AI, OpenAI API, Azure OpenAI API, Anthropic API, Groq, Ollama

**RAG, Knowledge Retrieval & Evaluation:** RAG Pipelines, Retrieval-Augmented Generation, Semantic Search, Vector Databases (ChromaDB), Embeddings, TF-IDF Retrieval, Knowledge Retrieval Systems, Document Intelligence, OCR, Confidence Scoring, LLMOps, MLOps, Model Evaluation, LLM-as-Judge, Hallucination Detection, CI-Integrated Regression Gates, Agent Observability, Distributed Tracing

**Languages, Frameworks & Enterprise Integration:** Python (9yr), TypeScript, JavaScript, SQL, FastAPI, Flask, REST APIs, Enterprise Systems Integration, Async Processing (RabbitMQ), Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, NoSQL, SQLite, Redis, Data Pipelines, ETL

**Cloud, DevOps & Data:** AWS (S3, EC2, RDS, Glue), Azure (App Service, Azure SQL, cloud services), Google Cloud Storage (GCP), Docker, Kubernetes, CI/CD (GitHub Actions), Performance Optimization, Query Optimization (200+ SQL queries, 30% gain), Pandas, NumPy, Tableau, Power BI, Git, Linux, Agile/Scrum, Stakeholder Management, Technical Mentoring

## EXPERIENCE

### Magizhs Tech Solution

May 2025 -- Aug 2025

**AI/ML Engineer -- Agentic AI Delivery**

- Sole technical owner of a production AI system built 0--1: designed LangGraph-based autonomous agent workflow, built RAG retrieval layer over FAA regulatory corpus with compliance guardrails, implemented LLMOps evaluation pipeline and CI validation gate, deployed to production with measurable compliance outcomes. End-to-end: from stakeholder requirements through agentic design, RAG architecture, guardrail implementation, and production delivery.

### George Mason University

Aug 2024 -- May 2026

**Graduate Teaching Assistant & Research Associate**

- Built multi-source ML prediction pipelines (Python, R) on AWS and Azure across 190+ countries (R2=0.977); applied Isolation Forest anomaly detection across 3 research pipelines, surfacing data quality failures manual review missed; designed Tableau, ggplot2, and Quarto dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems.

### Entrata -- Multifamily Real Estate SaaS

Sep 2018 -- Aug 2024

**Senior Software Engineer -- Enterprise AI & Production Systems · Lehi, UT (Remote)**

- **Enterprise production systems at scale** -- shipped 20+ full-stack modules (Python, JavaScript, PostgreSQL) across 120+ Agile sprints for 100K+ MAU; owned a 7-year data pipeline (few thousand federal source rows -- 180,000 domain records) powering eligibility, financial, and compliance calculations with zero violations; built Python/Bash automation eliminating 80% of manual operational workload; led cross-functional delivery with compliance, operations, and business stakeholders.
- **Cloud infrastructure & enterprise integration** -- led AWS S3 document migration: inventoried all document types, mapped legacy records to framework contracts, implemented Libsodium-encrypted ETL pipelines with 100% PHI compliance and zero security incidents; integrated enterprise workflows with REST APIs, RabbitMQ async processing (retry logic, idempotency guarantees), and multi-tenant database systems; 200+ SQL query optimizations (30% performance gain).
- **AI systems & developer productivity** -- built internal investigation and debugging tools cutting team triage time 40%, sustaining bug count under 10 for 6+ consecutive months; Docker microservices redesign (50% scalability improvement, 40% deployment time reduction); GitHub Actions CI/CD; mentored 5+ engineers; 300+ peer code reviews.

### Entrata India -- Pune

Jan 2017 -- Sep 2018

**Associate Software Engineer -- Python · PostgreSQL · AWS**

- Built property management and compliance modules (Python, PostgreSQL); 1.6M HUD/Tax Credit record pipeline; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific regulatory document templates.

## PROJECTS

### Leasel: Agentic AI Platform for Enterprise Compliance Operations

Generative AI · LangGraph · RAG · HITL · Responsible AI · Enterprise Deployment

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Enterprise AI platform transforming how compliance teams investigate issues and process regulatory documents -- a working proof of production agentic AI in a regulated domain. **Autonomous agent workflow:** 8-agent LangGraph StateGraph (Triage -- Retrieval -- Configuration -- Historical Incident -- Structured Data -- Reasoning -- Validation -- Human Escalation) with conditional routing, HITL escalation on low-confidence paths, and deterministic safety boundaries enforced in code. **RAG & knowledge retrieval:** hand-rolled TF-IDF retrieval with section-aware chunking, OCR fallback, 100% citation correctness on eval set. **Responsible AI:** LLM reasons and extracts; human or deterministic logic makes every final regulated decision -- guardrail enforced in code, not just in prompts. **Evaluation:** 93.6% precision, 87.9% F1 across 1,240 test cases; LLMOps regression gate via AgentProbe CI pipeline.

### AgentProbe: Open-Source LLMOps & AI Evaluation Toolkit

LLMOps · Model Evaluation · LangGraph · CI Gate · Multi-Provider · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic / OpenAI / Groq / Ollama APIs | github.com/venkatalakshmiparimala/AgentProbe

Production LLMOps toolkit for monitoring and optimizing deployed AI applications -- evaluates accuracy, hallucination rate, retrieval quality, latency, and token efficiency across LLM providers. **Evaluation infrastructure:** 5-metric LLM-as-judge engine, pass@k multi-trial sampling for non-deterministic output variance, CI-integrated regression gate (GitHub Actions) blocking quality degradation before deployment. **Agentic layer:** 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with ChromaDB semantic retrieval and structured observability across all agent steps. **Platform:** 15-endpoint FastAPI backend, React.js + TypeScript frontend, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) -- 100% local, zero cost. Used to evaluate Leasel across 1,240+ compliance test cases with zero regressions.

### AI on FHIR: Enterprise AI Application on AWS & Azure -- Regulated Domain

Generative AI · REST APIs · AWS · Azure · HIPAA · Full Stack · Enterprise Integration

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · Azure · PostgreSQL · REST APIs · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

Enterprise AI application integrating generative AI capabilities with regulated healthcare systems -- LLM-powered natural language query translation (EN/ES) into structured FHIR REST API calls; HIPAA-compliant RBAC with full audit trail; AWS Lambda and Azure cloud services; Docker containerization; observability stack with compliance tracing. Demonstrates AI integration with enterprise APIs, cloud platforms, and data sources in a high-stakes regulated environment -- achieved ~60% reduction in manual clinical retrieval time.

## EDUCATION

**M.S. Data Analytics Engineering --** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering --** JNTU Anantapur — 2013--2017